Stat 705 Final Project

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I Introduction

Executive Summary

This report seeks to replicate the basic elements of a paper by Siem Jan Koopman and Rutger Lit on modeling match results in the English Premier League [1]. It will discuss the other ideas posed by the authors and the results found. The paper, revised in 2013, implements a dynamic bivariate Poisson model and incorporates several advanced statistical methods to simulate and forecast outcomes. The *bivariate* Poisson distribution is an extension of a *double* Poisson distribution, which is itself simply the product of two independent Poisson distributions. The bivariate case involves a dependence parameter that captures relationship between the random variables. In our case, this is the correlation between goals scored by the home team and goals scored by the away team. The means of said distribution are taken to be the goal scoring intensities of the two teams at play, each of which are a function of their respective attacking and defending strengths. Notable methods are the formatting of the model with a state-space representation, and computation of likelihoods via Markov chain Monte Carlo (importance sampling) techniques. The model is assessed by splitting the data into a training and a test set - the last two seasons used as the latter to determine predictive performance. Additionally, the authors explore the utility of the model as it relates to sports betting, showing that it is capable of guiding winning strategies.

Literature Review

Some of the earliest work on the topic is found in Maher (1982) [10], which used a double Poisson distribution with means representing attack and defense strengths for each specific team. Koopman and Lit note that this paper did not allow either of these parameters to vary over time - a primary motivation of their subsequent methodology. In 1997, Dixon and Coles [3] implemented the same model but with the inclusion of a dependence parameter which captured the correlation between home and away match scores, concluding it is reasonable to assume independence between said scores aside from 0-0, 1-0, 0-1, and 1-1 results. Their inclusion of a weighting function to reduce the effect of results from the distant past was noted as well. Karlis and Ntzoufras (2003) [7] concluded that a small value for the dependence parameter provides more accurate simulations of draws, a notoriously difficult thing to model. They also did not allow attack or defense strength to vary over time. Rue and Salvesen (2000) [12] developed the work from Dixon and Coles (1997) [3] by using Markov chain Monte Carlo methods within a dynamic generalized model to analyze how attack and defense strength may change over time. Notably, they reduced any instances of a team scoring more than 5 goals in a match to just 5, arguing any successive goals do not provide significant information on a team's strengths. Owen (2011) [11] continued this work even further, utilizing the same framework, finding that attack and defense strengths and the dependence between home and away scores can be more effectively analyzed in *discrete time.* Crowder *et al.* (2002) [2] is the first paper cited by the authors to apply the problem to a non-Gaussian state-space model, allowing for the time variance of attack and defense strengths. They resorted to approximate methods for parameter estimation due to the computationally expensiveness of such a task at the time.

The noted improvements and changes implemented by Koopman and Lit are that they verified their model with a larger data set than Owen (2011) [11], and did not resort to approximate methods to compute parameter estimates unlike Crowder *et al.* (2002) [2]. Instead, maximum likelihood methods were utilized to routinize the process. Additionally, they allowed for the attack and defense strengths of teams to vary over time, and concluded that goals scored do not have to be truncated to 5. They also explored some additional model variants which are detailed in Section III. It is worth noting that other studies have been done across other sports leagues. This includes but is not limited to: Glickman and Stern (1998) [6], Fahrmeir and Tutz (1994) [5], and Knorr-Held (2000) [9]. These analyzed the American Football League, the German *Bundesliga*, and the (American) National Basketball Association, respectively.

II Goal and Rationale

The primary inference task is to predict the results of EPL matches, which can be done out of pure interest or to inform gambling decisions. Sports betting is a rapidly-growing worldwide industry, with 11 Premier League clubs in the 2024/25 season being sponsored by gambling companies. The total value of these sponsorships is 125.5 million GBP, an increase of 66% from the 2023/2024 season [14]. Numbers of similar or even greater magnitude can be seen in sports leagues across the world, especially in the United States where the online sports betting industry is valued over 16 billion USD [15]. As an extremely lucrative industry, tools for modeling the basis of said industry can prove useful and potentially profitable. The authors assert that their model can "produce a significant positive return over the bookmaker's odds." Sub-objectives include determining the extent to which team qualities change over time, insight into the overall effect of summer and winter breaks on goal-scoring propensity (discussed in more detail in the next section), a quantification and significance test on the effect of playing on one's home field, and the interdependence between home scores and away scores. Also of use from the paper is a comparison across models as it relates to predictive performance. The main outcome of this report is to gain an understanding on methods used to simulate from complicated models - specifically via Monte Carlo importance sampling.

III Methods - Statistical Modeling Framework

The following section will summarize the fundamentals of the methods used and the formulation of specific models. As mentioned, the authors incorporated or improved upon numerous elements from the previously discussed papers, seeking to resolve potential weaknesses and provide more meaningful results.

Bivariate Poisson Model

The backbone of the paper is of course the bivariate Poisson distribution, which has the density function

$$p_{BP}(X,Y;\lambda_x,\lambda_y,\gamma) = e^{-(\lambda_x+\lambda_y+\gamma)}\frac{\lambda_x^X}{X!}\frac{\lambda_y^Y}{Y!}\sum_{k=0}^{min(X,Y)}\binom{X}{k}\binom{Y}{k}k!(\frac{\gamma}{\lambda_x\lambda_y})^k$$

The above can be written in shorthand notation with $(X,Y) \sim BP(\lambda_x,\lambda_y,\gamma)$. The random variables are of course X_{it} and Y_{jt}, representing the number of goals scored by the ith (home) and jth (away) team in week $t = 1,...,n$, where n is the number of weeks in the data set. Here, $n = 342$ since each of the 20 clubs play 38 games per season and there are 9 seasons. Each week sees 10 matches played (since the teams all play each other), and thus a single season consists of 38 weeks (as a double-round-robin format). The intensities for X and Y are then λ_x and λ_y, respectively, while γ represents the covariance between the variables. The means and variances of X and Y are $\mathbb{E}(X) = Var(X) = \lambda_x + \gamma$, and $\mathbb{E}(Y) = Var(Y) = \lambda_y + \gamma$, respectively.

Dynamization of Goal-Scoring Intensity Parameters

The strengths of attack and defense for the teams were allowed to change over time, operating under the assumption that they must not be constant due to various factors. The authors also included an additional parameter δ to encapsulate the effect of home-ground advantage, which is assumed to be the same for all teams (initially) and constant over time. Define the following:

$$\lambda_{x,ijt} = e^{\delta+\alpha_{it}-\beta_{jt}}, \quad \lambda_{y,ijt} = e^{\alpha_{jt}-\beta_{it}}$$

where α_{it} represents the strength of attack for team i in week t, and β_{it} represents the strength of defense for said team in week t. j represents the opponent of team i in week t. Naturally, the goal-scoring intensity of team i is expected to be the difference between their own attack strength and the defense strength of the opposing team. Of course, the away team cannot have a home ground advantage, and thus their intensity does not include δ. The parameters α and β are assumed fully independent and unique across all teams, and are allowed to change over time via specification as auto-regressive processes. See the following for $\kappa = \alpha, \beta$:

$$\kappa_{it} = \mu_{\kappa,i} + \phi_{\kappa,i}\kappa_{i,t-1} + \eta_{\kappa,it}$$

Here, the μ's are unknown constants, with the ϕ's representing the auto-regressive coefficients. Random error, or "shocks" to the attack and defense strengths of the teams are captured by the respective η's and are assumed to be iid normally distributed with mean 0 and variance $\sigma^2_{\kappa,i}$. The initial conditions for the processes were based on the means and variances of the respective unconditional distributions of the parameters: $\mathbb{E}(\kappa_{it}) = \mu_{\kappa,i}/(1-\phi_{\kappa,i})$ and $Var(\kappa_{it}) = \sigma^2_{\kappa,i}/(1-\phi^2_{\kappa,i})$. Importantly, the authors decided to have the auto-regressive coefficients be constant among all teams in order to avoid a prohibitive increase in the number of parameters to estimate. In any case, the above specification allows for the primary goal-scoring aspects of the team to vary over time, which will lead to better model performance as teams usually change in quality over time, especially over a long duration.

Additional Extensions

Diagonal Inflation Adjustment

The authors explored a few other modeling avenues to improve fit and explore other potential factors influencing match results. The first was to correct for the underestimation of draws, motivated by the findings of Dixon and Coles (1997) [3] which used a diagonal inflation adjustment to adjust the bivariate density by a factor of $\pi_{\lambda_x,\lambda_y}(X,Y)$, giving more mass to final scores of (0,0), (0,1), (1,0), and (1,1). The exact specification for this can be found in the appendix.

Summer and Winter Breaks

These breaks are used by teams as the primary time to sign or release players, thereby improving or perhaps worsening their overall team quality. As such, the breaks can have a real effect on the attack and defense strengths of teams. This effect was captured by altering the distribution of $\eta_{\kappa,it}$, $\kappa = \alpha, \beta$ to

$$\eta_{\kappa,it} \sim ind.N(0, \sigma^2_{\kappa,i} + \sigma^2_{\kappa,S}\tau_S(t) + \sigma^2_{\kappa,W}\tau_W(t))$$

$\tau_S(t)$ and $\tau_W(t)$ are indicator variables which are set to 1 at the end of summer and winter breaks, respectively, and are 0 at all other times. This adds extra random "shocks" to attack and defense strengths of teams during breaks, which could capture the addition or loss of players that may have an effect on the goal-scoring propensity of the team.

Home Ground Advantage

Thirdly, Koopman and Lit considered that the home ground advantage may not be the same for all teams. Instead of having a specific advantage for each team, which would have led to too many parameters and slowed down the estimation process, they pooled home ground advantage coefficients for specific groups of teams. Namely, they considered the advantage for the classically well-performing ("elite") clubs Arsenal, Chelsea, Liverpool, Manchester City, and Manchester United to be different from the rest of the teams in the league. The effect of this addition, along with the others discussed, is analyzed later.

State-Space Representation

A description of the *full* state-space representation of the model is described in this section, though the actual implementation was far simpler due to keeping ϕ_α, ϕ_β the same for all teams. The dynamic model for the result of home team i against team j in week t is $(X_{it}, Y_{jt}) \sim BP(\lambda_{x,ijt}, \lambda_{y,ijt}, \gamma)$. The first two parameters use the link functions

$$\lambda_{x,ijt} = s_{x,ij}(z_t) = e^{\delta + w_{ij} z_t}$$

$$\lambda_{y,ijt} = s_{y,ij}(z_t) = e^{w_{ij} z_t}$$

Here, the "state vector" z_t contains the attacking and defending strengths of all teams at time t:

$$z_t = (\alpha_{1t}, ..., \alpha_{Jt}, \beta_{1t}, ..., \beta_{Jt})'$$

J represents the total number of teams, and $t = 1, ..., n$ is the t^{th} week in the data set. w_{ij} is a selection vector which chooses the appropriate α's and β's from z_t corresponding to the correct teams.
The authors write the linear dynamic process as $z_t = \mu + \Phi z_{t-1} + \eta_t$, where $\eta_t \sim N(0, H)$ where H is the covariance matrix. The components of z_t, along with the formulation of H, are as follows:

$$\mu_{(2J\times 1)} = (\mu_{\alpha,1}, ..., \mu_{\alpha,J}, \mu_{\beta,1}, ..., \mu_{\beta,J})$$

$$\Phi_{(2J\times 2J)} = diag(\phi_{\alpha,1}, ..., \phi_{\alpha,J}, \phi_{\beta,1}, ..., \phi_{\beta,J})$$

$$H_{(2J\times 2J)} = diag(\sigma^2_{\alpha,1}, ..., \sigma^2_{\alpha,J}, \sigma^2_{\beta,1}, ..., \sigma^2_{\beta,J})$$

μ represents the base-level attacking and defending propensities for each of the J teams, Φ contains the auto-regressive coefficients, and H contains the variances governing the random perturbations affecting each α and β. Since the base-level effects are captured by the unknown initial state vector z_1, the remaining parameters can be collected in $\psi = (\phi', h', \delta, \gamma)$, with δ and γ of representing the home-ground advantage and the interdependence between X and Y, respectively. This serves as the *primary parameter vector if interest*. To simplify things, the authors pooled together the unknown coefficients into smaller sets of parameters - i.e. having the auto-regressive coefficients and their variances be the same among all teams. This dramatically reduces the dimension of the above equations, and improves computational feasibility.

Likelihood Evaluation

The final method to summarize is the manner in which the authors did maximum likelihood estimation for the parameters of the model. Since we have a complex framework, the calculation is not trivial and requires advanced methods. They implement Monte Carlo maximum likelihood via importance sampling.
Collecting all the goals scored in week t into vector y_t, the resulting observation density of y_t given z_t is

$$p(y_t|z_t;\psi) = \prod_{k=1}^{J/2} f(\lambda_{x,ijt}, \lambda_{y,ijt}, \gamma)$$

where f is the bivariate Poisson density, and the product is taken to $J/2$ because there are $J/2$ match results in each week.

Denoting $y = (y_1', ..., y_n')'$ and $z = (z_1', ..., z_n')'$, we have $p(y|z;\psi) = \prod_{k=1}^{n} p(y_t|z_t;\psi)$, which implies that the scores from week-to-week are conditionally independent given strengths of attacks and defense, γ, and δ. Note also that the $p(y, z; \psi)$ can be written as $p(y|z;\psi)p(z;\psi)$, where $p(z;\psi) = p(z_1;\psi)\prod_{i=2}^{n} p(z_t|z_1, ..., z_{t-1};\psi)$ Since z_t is an auto-regressive process, the computation of $p(z_t|z_1, ..., z_{t-1};\psi)$ is simplified.
The likelihood (objective) function for y is given by

$$l(\psi) = p(y;\psi) = \int p(y, z;\psi)dz = \int p(y|z;\psi)p(z;\psi)dz$$

which can be expanded into

$$\int \prod_{t=1}^{n} \prod_{k=1}^{J/2} f(\lambda_{x,ijt}, \lambda_{y,ijt}, \gamma)p(z_1;\psi) \prod_{i=2}^{n} p(z_t|z_1, ..., z_{t-1};\psi)dz$$

Numerical methods are required because an analytical solution to this integral is not available. These methods are the aforementioned Monte Carlo simulation techniques. Importance sampling provides an efficient and effective approach, and is discussed in detail by the following. The methods are developed by Shephard and Pitt (1997) [13] and Durbin and Koopman (1997) [4]. The shorthand used for these techniques is SPDK.

Importance Sampling - Set-Up

We will notate an appropriate approximating linear Gaussian model with $g(y, z;\psi)$, which can be re-written as $g(z|y;\psi)g(y;\psi)$, which is of similar form to the likelihood $l(\psi)$. Denote $l_g(\psi) = g(y;\psi)$, and note that it can be expressed as

$$l_g(\psi) = g(y;\psi) = \frac{g(y, z;\psi)}{g(z|y;\psi)} = \frac{g(y|z;\psi)p(z;\psi)}{g(z|y;\psi)}$$

since $p(z;\psi) \equiv g(z;\psi)$. If we substitute $p(z;\psi) = g(y;\psi)g(z|y;\psi)/g(y|z;\psi)$ into $l(\psi)$, we have

$$l(\psi) = g(y;\psi)\int \frac{p(y|z;\psi)}{g(y|z;\psi)} g(z|y;\psi)dz = l_g(\psi)\mathbb{E}_g \frac{p(y|z;\psi)}{g(y|z;\psi)}$$

since y_t observations are independent for given z_t for each time point $t = 1, ..., n$, $g(y|z;\psi) = \prod_{t=1}^{n} g(y_t|z_t;\psi)$.

Importance Sampling - Construction of Approximating Model

The next step is to construct an appropriate approximating model, which is given by the following according to SPDK:

$$g(y, z;\psi) = g(y|z;\psi)g(z;\psi) = g(z;\psi)\prod_{t=1}^{n} g(y_t|z_t;\psi)$$

with $g(z;\psi)$ representing the density of $z_t = \mu + \Phi z_{t-1} + \eta_t$, the dynamic state process. $g(y_t|z_t;\psi)$ is represented by independent distributions

$$N(a_t\delta + W_t z_t + c_t, V_t)$$

a_t is 1 when goals scored in y_t is by the home team, 0 when by the away team. W_t selects the α's and β's from z_t according to the teams at play, though these will be the same for all teams in this analysis. c_t is a mean correction, which along with variance V_t is chosen so that the derivatives of the log densities of the true and approximate likelihoods are equivalent. These solutions are derived in the authors' online appendix and are written below:

$$V_t = -W_t\ddot{p}_t^{-1}W_t', \quad c_t = y_t - a_t\delta - W_t(z_t + \ddot{p}_t^{-1}\dot{p}_t(z_t))$$

Since the equations depend on time t, they must be solved for iteratively. Analytic expressions for the derivatives $\dot{p}$ and $\ddot{p}$ are given in the online appendix, and are determined by the bivariate poisson parameters (and thus the time-varying strengths of attack and defense).
After putting everything together, a simple estimate of the function is essentially the "average" function (of ψ) over M Monte Carlo replications:

$$\hat{l}(\psi) = l_g(\psi)\frac{1}{M}\sum_{i=1}^{M} w_i, \quad w_i = \frac{p(y|z^{(i)};\psi)}{g(y|z^{(i)};\psi)}, \quad z^{(i)} \sim g(z|y;\psi)$$

The estimated $\hat{l}(\psi)$ is then maximized by multivariate Newton-Raphson to compute parameter estimates.

Model Specifications

The authors tested seven model specifications, which are listed below.
a) the basic model with parameters $\psi = (\phi_\alpha, \phi_\beta, \sigma_\alpha^2, \sigma_\beta^2, \delta, \gamma)$;
b) the basic model with fixed coefficients for the auto-regressive processes, meaning the parameter vector consists only of γ and δ;
c) the basic model with $\gamma = 0$, i.e. a double Poisson distribution for match results;
d) the basic model with time-varying and team-specific γ, given by $\gamma_{ijt} = \gamma^*(\lambda_{x,ijt}\lambda_{y,ijt})^{1/2}$, where $\gamma^* \geq 0$ is a scaling coefficient replacing γ in the parameter vector ψ. Since the dependence coefficient itself depends on the strength of attack and defense, both of which vary over time in our set-up, it is logical to allow it to vary over time as well;
e) the model adjusted with the diagonal inflation to avoid underestimation of low-scoring results and specifically draws;
f) the basic model with the inclusion of additional shocks to the change in attacking or defending strength over time, caused by summer and winter breaks;
g) the basic model with home ground advantage parameters for the "Elite" teams listed previously and the remaining teams. This gives us two parameters to estimate, δ_1 for the elite group, δ_2 for the rest of the league.

IV Results

Model Fitting

Parameter Estimation

The code in the report will be supported by the "bivpois" package developed by Michail Tsagris in 2023 [16] and additional example code using an older package is provided by Karlis and Ntzoufras (2005) [8]. The basic model from the original paper was implemented, while the more advanced variants are discussed.

The Table 1 summarizes the Monte Carlo parameter estimates for the basic model along with the log-likelihood for $M = 50, 200, 1000$ simulated paths. The results are similar regardless of the value of M, an observation that applies to each parameter. ϕ_α and ϕ_β being close to 1 suggests the strengths are persistent and behave as random-walk processes. We expect to see greater changes over longer periods of time - since there are 38 weeks, we calculate $0.9985^{38} = 0.94$ and $0.9992^{38} = 0.97$. This implies that the effect is slightly larger from season to season rather than week to week. The estimates for σ_α^2 and σ_β^2 are both very small, suggesting attack and defense strengths vary smoothly over time. That is, there are not *major* fluctuations for either facet, like sudden jumps or collapses in quality. The γ estimate captures a slight positive relationship between away goals and home goals, meaning that if one team scores more goals, the other is expected to increase their output slightly. The δ estimate being fairly large suggests that there is a tangible advantage in goal-scoring when playing at home. While the authors do not explicitly comment on the statistical significance of these parameters, they all have very small standard errors, and would thus

Table 1. Estimates of parameter vector ψ†

ψ	*Results for* $M = 50$	*Results for* $M = 200$	*Results for* $M = 1000$
ϕ_α	0.9985 (0.00044)	0.9985 (0.00044)	0.9985 (0.00044)
ϕ_β	0.9992 (0.00027)	0.9992 (0.00027)	0.9992 (0.00027)
σ^2_α	0.000205 (2.20×10^{-5})	0.000206 (2.27×10^{-5})	0.000206 (2.28×10^{-5})
σ^2_β	0.000141 (2.05×10^{-5})	0.000143 (2.02×10^{-5})	0.000143 (2.02×10^{-5})
δ	0.3662 (0.0196)	0.3643 (0.0269)	0.3641 (0.0252)
γ	0.0966 (0.0232)	0.0966 (0.0232)	0.0966 (0.0232)
$\hat{l}(\psi)$	−9608.56	−9608.38	−9608.38

†The table reports the Monte Carlo estimates for the parameter vector ψ together with the value of the maximized log-likelihood value for different numbers of simulated paths $M = 50, 200, 1000$. The Monte Carlo estimates of the standard errors are given below the estimates and between parentheses. The data set that was used for estimation covers seven seasons of the English Premier League (from 2003–2004 to 2009–2010).

Figure 1: Koopman and Lit Table 1

be significant under many test levels.

The code in the appendix includes a useful data visualization and summary function. Named match_results, the user can select a team and a subset of seasons to get basic summary statistics of said team's data over said time frame, along with plots of all the goals. This can be used to recreate the goal plots from the original paper, and arguably with better visuals. Examples of these are included in the appendix of this report. There are also functions which re-format and plot the data in additional useful ways.

Forecasting and Model Comparisons

This section analyzes the results provided by the authors for the various models implemented. Due to computational complexity, this report was limited to an attempt at the most basic model.

Table 2 above summarizes the comparisons between the models implemented by Koopman and Lit as it relates to forecasting performance. The loss function was extremely similar for all models aside from model b). The *Diebold-Mariano* test in the following column comes up with a statistic for comparing predictive performance - we see that each model other than b) was insignificant when compared to the base model a). Specifically, b) was significantly worse than a), and is thus expected to be worse than all of the other models. This is strong evidence for the argument that attack and defense strengths *do* vary over time. Furthermore, model e) with the diagonal inflation adjustment is the only one to out-perform a), albeit insignificantly. Looking at the models individually, with the "t-test" column we see that the alterations made by models c), d), and f) were significantly different from 0. These tell us that: c) there is significant dependence between home goals and away goals, d) this dependence holds when varying over time, f) summer and winter breaks have a significant effect on the changing of attack and defense strength over time. The last row tells us that δ_1 is not significantly different from δ_2, meaning there is no additional advantage for the "elite" teams when playing on their home grounds.

Betting Strategy

The last significant portion of the paper covers the application of the model to betting strategy. This is an extra verification step, using the last two seasons of the data. By plugging in the estimated parameter values

Table 2. Model comparisons: in-sample and out-of-sample results†

Model	*Model specification*	*Number of parameters*	H_0	*Likelihood ratio test*	*t-test*	*Squared loss*	*Diebold–Mariano test*
(a)	Basic model	6				2087.10	
(b)	Basic model with time invariant signals	6/2		123.04‡		2190.80	−3.67‡
(c)	Basic model with no dependence	6/5	$\gamma = 0$		4.16‡	2087.90	−0.62
(d)	Basic model with time varying dependence	6/5	$\gamma^* = 0$		3.84‡	2088.60	−1.51
(e)	Diagonal inflation model	7/6	$\omega = 0$		1.85	2086.70	0.91
(f)	Summer break for signals	7/6	$\sigma^2_{\kappa,S} = 0$		2.84‡	2098.50	−1.75
(g)	Two home ground advantages	7/6	$\delta_1 = \delta_2$		0.35	2089.00	−1.08

†We compare the in-sample fit and out-of-sample forecasting accuracy for seven model specifications. The number of parameters (p_1 / p_0) is given for each model (p_1) and for the model under the null hypothesis H_0 (p_0); see Section 3.5 for further details. The in-sample results are based on seven seasons of the English Premier League (from 2003–2004 to 2009–2010). The *t*-tests are computed and presented for the hypotheses with a single restriction whereas the likelihood ratio test is presented for the multiple restriction in model (b). The out-of-sample results are based on the two seasons 2010–2011 and 2011–2012. The squared loss functions and the Diebold–Mariano tests are based on one-step-ahead forecasts from a rolling window sample.
‡Significance at the 5% level of significance.

Figure 2: Koopman and Lit Table 2

into the bivariate Poisson model, one can compute the probabilities of each outcome in a match.
To illustrate, the authors discuss the example of Aston Villa's home match against West Ham in the 2010-2011 season. The forecast parameter values were $\lambda_{x,ij,t+1} = 1.7272$ and $\lambda_{y,ij,t+1} = 0.8127$, giving a win probability of 0.591 to Aston Villa, 0.174 for West Ham, and a 0.235 probability of a draw. The odds for the game were 1.96 for Aston Villa to win at home, 4.03 for a West Ham win, and 3.30 for a draw. We calculate the expected value of a unit bet as

$$EV(A) = P(A)[odds(A) - 1] - P(A^C) = P(A)odds(A) - 1,$$

with A representing a win, loss, or draw events of the home team, $P(A)$ the probability of A, and $odds(A)$ the bookmaker's odds for A. Plugging in the previous values, the only positive expected return is a bet on the home team winning. The simplest strategy would be to take bets where $EV(A) > 0$, but a less risky strategy involves using a benchmarking threshold $\tau > 0$. That is, only place a bet if the expected value exceeds this benchmark. Long shot events are also considered, which are low probability events with very high odds (taken to be 7 by Koopman and Lit).
To determine odds, the authors took the average of those provided by 28-40 bookmakers in the 2010-11 and 2011-12 seasons. Summing the reciprocals of the bookmakers odds and subtracting 1 gives the profit of the bookmaker (6.1% in the example above, and 7% on average, meaning the expected profit of a unit bet is -0.07). This provides a benchmark, as the model and aforementioned strategy must out-perform this 7% advantage *and* generate a positive overall return. *Bootstrap methods* are applied to estimate the profit of the strategy. It was found that $\tau > 0.12$ is where positive returns begin to be obtained, though the number of betting opportunities naturally decreases as the threshold rises.

The plots above were also created by the original authors, and plot τ on the horizontal axis. Since τ is a benchmark for the expected value of a bet, the plots are logical. For the top plot, we naturally expect the mean return to increase with the value of τ. This does not happen smoothly, though, since there are a number of "long shots" in the data. The bottom plot is also intuitive in that the number of value bets will monotonically decrease as τ increases, because the bettor has a higher benchmark for what they consider to be a "value bet". Users can decide a satisfactory τ and expected return for their own purposes - both of which are easily guided by the models. The ability for the models to compute result probabilities is intuitive and obviously useful as it relates to gambling strategy.

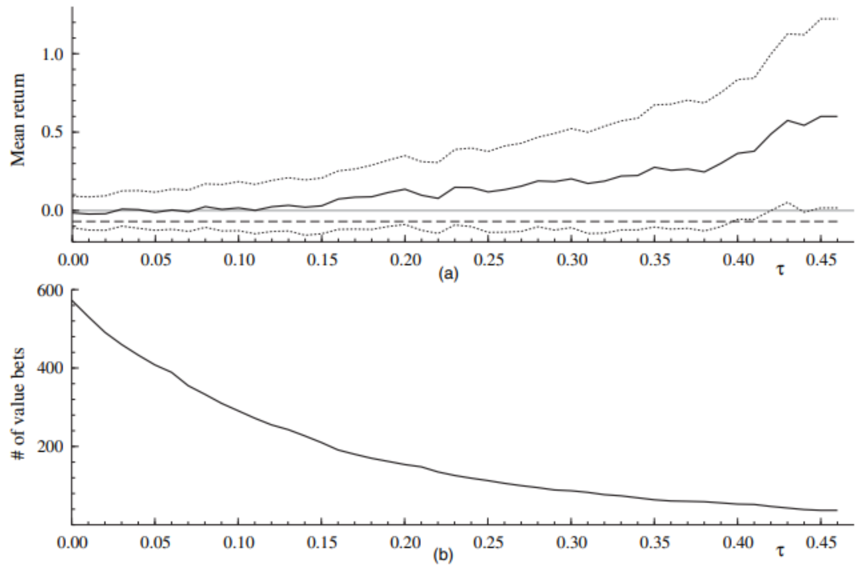

Fig. 3. Returns of a betting strategy for the 2010–2011 and 2011–2012 seasons of the English Premier League: (a) average return from betting on match outcomes by using our strategy for various values of the threshold τ (———), average return under random betting which we have established at -0.07 (– – –) and 90% bootstrap confidence intervals (· · · · · ·); (b) number of quality bets for various values of τ out of the 760 betting opportunities in the two seasons

Figure 3: Koopman and Lit Fig. 3

V Discussion

If we consider Table 2 from the original paper, it could be said that the basic model is still the most desirable. Although models c), d), and f) have significant additional parameters, the predictive performance is not significantly *better* than the base model. It may be more attractive to choose the simplest model for ease of implementation. The first key takeaway is that the assumption of team strengths evolving over time is a good one, as Table 1 gives significant estimates for the auto-regressive coefficients. The *bivariate* Poisson framework is also appropriate based on the significant result for γ in the same table, along with the random fluctuations of attack/defense strengths represented by the σ^2's. The base model implies the significant effect of a home-ground advantage with the low standard error of the δ estimate. Also of note is that the parameter estimates are quite similar regardless of the number of Monte Carlo iterates, which suggests future procedures may be able to save on computing time by reducing said number. This time could also be allocated toward other inference tasks, such as those discussed next.

An additional inferential avenue would be a deeper analysis of the effect of home ground advantage - perhaps a model which allows said effect to vary over time. It seems a logical assumption that if the quality of a team changes, so too would any other advantage they may possess. This could also be explained by lower-quality teams potentially having less consistent fan support, though this may be difficult to capture mathematically and may not apply to every team. In a more general sense, it may be useful to incorporate other elements of the sport. While goal-scoring and defending are obviously the most pertinent aspects, other statistics are available which can offer some explanatory power for team performance. These include possession percentage, total shots, shots on target, passing percentage, or any of a plethora of advanced football statistics.

Another key assumption in Koopman and Lit's work is that the dynamic processes determining attacking and defending strength are independent. It may be plausible that this is not always the case, as the midfielder position often has an impact on *both* aspects simultaneously. The signing or loss of a strong midfielder could have a tangible effect on both the offensive *and* defensive prowess of a team. This idea may also bring about the idea of a model which somehow accounts for the individual effect of *each player* on a team. However, such a model would be very complicated and perhaps intractable. It may also be hindered by over-parameterization, which would reduce the utility of the model as a whole. In any case, these additional methods are worth exploring as means of improving the model, which could be assessed in the same predictive

way done by Koopman and Lit.
These methods, both the "basic" ones and the additional ones discussed above, could be applied to other sports as well. There great deal of analytical literature on baseball, which is perhaps the most comparable along with hockey due to their scoring magnitudes. American football and basketball may prove more challenging due to the scoring complexity of the former and scoring magnitude of the latter. However, the authors themselves described their methods as a "novelty in the statistical time series analysis of match results in team sports". The relevancy of the dynamic attacking and defending strengths of teams to other sports not being lost, it is sensible to conclude that said methods have great potential in external applications.

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Main

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(Online Appendix linked at bottom of page linked above)

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Appendix

Equations

Diagonal inflation adjustment: $\pi_{\lambda_x,\lambda_y}(X,Y)$

$$\pi_{\lambda_x,\lambda_y}(X,Y) = \begin{cases} 1+\lambda_x\lambda_y\omega & if(X,Y)=(0,0) \\ 1-\lambda_x\omega & if(X,Y)=(0,1) \\ 1-\lambda_y\omega & if(X,Y)=(1,0) \\ 1+\frac{\omega}{1+\gamma/\lambda_x\lambda_y} & if(X,Y)=(1,1) \\ 1 & otherwise \end{cases}$$

where ω determines how much probability mass is shifted.

R Code

```
# Load Packages, Set Up Data
library(ggplot2) # for plotting
library(tidyverse) # for data frame manipulation and pipe operator
library(dplyr)
library(dbplyr)
library(readxl) # for reading in data
library(bivpois) # for bivariate poisson density and random generation

epl_data <- read_excel("C:/Users/jacma/OneDrive/school work/UMD/Stat 705 Computational Stats/Final Proj

team_names <- read.csv("C:/Users/jacma/OneDrive/school work/UMD/Stat 705 Computational Stats/Final Proj

epl_data$Home_Name <- rep(NA,nrow(epl_data)) # new columns to see team names
epl_data$Away_Name <- rep(NA,nrow(epl_data))
epl_data$Result <- rep(NA,nrow(epl_data))

# for tabulating results
for (i in 1:nrow(epl_data)) {
  if (epl_data$FTR[i]==1) {epl_data$Result[i]<-"Home Win"}
  if (epl_data$FTR[i]==2) {epl_data$Result[i]<-"Away Win"}
  if (epl_data$FTR[i]==3) {epl_data$Result[i]<-"Draw"}
}

# easier to subset data, teams by name instead of number
for (i in 1:length(team_names$Number)){
  ind <- which(epl_data$HomeTeam==team_names$Number[i])
  epl_data$Home_Name[ind] <- team_names$Team[i]
  ind2 <- which(epl_data$AwayTeam==team_names$Number[i])
  epl_data$Away_Name[ind2] <- team_names$Team[i]
}

# Test / Training Data Split
epl_train <- epl_data[1:2660,] # first 7 seasons (2660 matches)
epl_test <- epl_data[2661:3420,] # last 2 seasons (760 matches)
```

```r
# clean up not-needed variables
rm(i,ind,ind2,team_names)
```

```r
# EDA
### Overall Summary and Distributions
# --> to analyze dispersion of primary variables of interest, can be used for
# basic assessment of predictive performance
frequencies <- list(Home_Goals=epl_data$FTHG %>% table(),
                    Away_Goals=epl_data$FTAG %>% table(),
                    Results=epl_data$Result %>% table() / nrow(epl_data),
                    Draw_Scores = subset(epl_data,epl_data$FTR==3)$FTAG %>% table())

### Specific Teams
# function takes in goal-scoring data, team, selected seasons, specified colors:
# --> provides summary stats of match results and scoring, plots all results
# --> also returns the data frame for the specific team seasons selected
match_results <- function(data,team,seasons,team_col="red",opp_col="lightgreen"){
  # for selecting correct dates
  years <- 2003:2012
  start_indices <- c(1,20,39,58,77,96,105,124,143)
  first_game <- start_indices[which(years==seasons[1])]
  last_game <- first_game + 19*(seasons[2] - seasons[1]) - 1

  home_data <- subset(data,data$Home_Name==team)
  home_data <- mutate(home_data, match_num = 1:nrow(home_data))
  home_data <- home_data[first_game:last_game,]
  all_home_goals <- ggplot(data=home_data) +
    geom_point(mapping=aes(x=match_num,y=FTAG,shape=Result),col=opp_col) +
    geom_point(mapping=aes(x=match_num,y=FTHG,shape=Result),col=team_col) +
    xlab("Match Number") + # improvement: dates here instead
    ylab("Goals Scored") +
    ggtitle(paste("All Goals Scored/Conceded by", team,"at Home",
                  seasons[1], "to",seasons[2],sep=" "),
            paste(team, "goals in", team_col, "opponent goals in", opp_col,sep=" "))

  home_results <- table(home_data$Result)

  # basic summary stats for the team and its opponents when said team is home
  team_home_stats <- data.frame(AvgGoals = mean(home_data$FTHG),
                                StdDevGoals = sd(home_data$FTHG),
                                AvgOdds = mean(home_data$`Home Odds`),
                                StdDevOdds = sd(home_data$`Home Odds`))

  opp_home_stats <- data.frame(AvgGoals = mean(home_data$FTAG),
                                StdDevGoals = sd(home_data$FTAG),
                                AvgOdds = mean(home_data$`Away Odds`),
                                StdDevOdds = sd(home_data$`Away Odds`))

  home_summary_stats <- rbind(Home_Team=team_home_stats,
                              Away_Team=opp_home_stats)

  rownames(home_summary_stats) <- c(team,"Opponent")
```

```r
  # collect everything
  home_summary <- list(Goals_Plot = all_home_goals,WLD = home_results,
                       Stats = home_summary_stats,Full_Results = home_data)

  away_data <- subset(data,data$Away_Name==team)
  away_data <- mutate(away_data, match_num = 1:nrow(away_data))
  away_data <- away_data[first_game:last_game,]
  all_away_goals <- ggplot(data=away_data) +
    geom_point(mapping=aes(x=match_num,y=FTHG,shape=Result),col=opp_col) +
    geom_point(mapping=aes(x=match_num,y=FTAG,shape=Result),col=team_col) +
    xlab("Match Number") + # improvement: dates here instead
    ylab("Goals Scored") +
    ggtitle(paste("All Goals Scored/Conceded by", team,"on the Road",
                  seasons[1], "to",seasons[2],sep=" "),
            paste(team, "goals in", team_col, "opponent goals in", opp_col,sep=" "))

  away_results <- table(away_data$Result)

  # basic summary stats for the team and its opponents when said team is away
  opp_away_stats <- data.frame(AvgGoals = mean(away_data$FTHG),
                               StdDevGoals = sd(away_data$FTHG),
                               AvgOdds = mean(away_data$`Home Odds`),
                               StdDevOdds = sd(away_data$`Home Odds`))

  team_away_stats <- data.frame(AvgGoals = mean(away_data$FTAG),
                               StdDevGoals = sd(away_data$FTAG),
                               AvgOdds = mean(away_data$`Away Odds`),
                               StdDevOdds = sd(away_data$`Away Odds`))

  away_summary_stats <- rbind(Home_Team=opp_away_stats,
                              Away_Team=team_away_stats)
  rownames(away_summary_stats) <- c("Opponent",team)

  # collect everything
  away_summary <- list(Goals_Plot = all_away_goals,WLD = away_results,
                       Stats = away_summary_stats,Full_Results = away_data)

  # return list containing all home information and away information
  return(list(Home=home_summary,Away=away_summary))
}
chelsea0711 <- match_results(epl_data,"Chelsea",c(2007,2011),
                             team_col = "blue",opp_col = "red")
chelsea0711$Home$Stats
chelsea0711$Home$Goals_Plot
chelsea0711$Away$Goals_Plot

# match_results(epl_data,"Newcastle",c(2008,2010),team_col="black",opp_col="gold")

team_all_data <- function(data){
  # takes all data rows and re-formats - separating by team
  # --> to combine home goals and away goals of team into one data-frame
  # similar work can be done for the Odds variables, to see if they change over
  # time for any specific teams
```

```r
  all_teams <- unique(data$Home_Name) %>% sort()
  team_data <- list()
  for (i in 1:length(all_teams)){
    team_data[[i]] <- subset(data, data$Home_Name==all_teams[i] |
                               data$Away_Name==all_teams[i])
    for (j in 1:nrow(team_data[[i]])){
      # create additional variables of interest, to be used in plotting function
      if (team_data[[i]]$Home_Name[j]==all_teams[i]){
        team_data[[i]]$Home[j] <- 1 # for plot coloring
        team_data[[i]]$`Team_Goals`[j] <- team_data[[i]]$FTHG[j]
        team_data[[i]]$`Team_Conceded`[j] <- team_data[[i]]$FTAG[j]
      }
      if (team_data[[i]]$Away_Name[j]==all_teams[i]){
        team_data[[i]]$Home[j] <- 0 # for plot coloring
        team_data[[i]]$`Team_Goals`[j] <- team_data[[i]]$FTAG[j]
        team_data[[i]]$`Team_Conceded`[j] <- team_data[[i]]$FTHG[j]
      }
      # column for points gained, use to color plots
      # could also aggregate for points gained by team in each season
      if(team_data[[i]]$Team_Goals[j] > team_data[[i]]$Team_Conceded[j]){
        team_data[[i]]$Pts[j]<-3}
      else if (team_data[[i]]$Team_Goals[j] == team_data[[i]]$Team_Conceded[j]){
        team_data[[i]]$Pts[j]<-1}
      else if (team_data[[i]]$Team_Goals[j] < team_data[[i]]$Team_Conceded[j]){
        team_data[[i]]$Pts[j]<-0}
    }
    team_data[[i]]$Home <- factor(team_data[[i]]$Home)
    team_data[[i]]$Pts <- factor(team_data[[i]]$Pts)

  }
  return(list(Team_List = all_teams, Data = team_data))
}
# contains list of data frames, each with all the results of every team playing
# at least one season in the Premier League from 2003-2011
team_results <- team_all_data(epl_data)
team_results$Team_List
team_results$Data[[1]]

all_goals_plotter <- function(all_info,team){
  # requires grouping function to be run first
  team_list <- all_info$Team_List
  index <- which(team_list==team)
  team_data <- all_info$Data[[index]]
  match_num <- 1:nrow(team_data)
  scored <- ggplot(data=team_data) + # x=year for goals in calendar year
    geom_col(mapping=aes(x=match_num,y=Team_Goals,fill=Pts)) +
    # x=Year for goals in calendar year, Month for months, Day for day of month
    # --> could reveal some interesting patterns, i.e. team scores more in certain
    # months, or for whatever reason more/less at the end/beginning of months
    ylab("Goals Scored") +
    xlab("Match Number") +
    ggtitle(paste("All Goals Scored by",team,sep=' '))
```

```r
  conceded <- ggplot(data=team_data) +
    geom_col(mapping=aes(x=match_num,y=Team_Conceded,fill=Pts)) +
    ylab("Goals Conceded") +
    xlab("Match Number") +
    ggtitle(paste("All Goals Conceded by",team,sep=' '))

  return(list(Scored=scored,Conceded=conceded))
}

all_goals_plotter(team_results,"Manchester City")
all_goals_plotter(team_results,"Manchester Utd")
### Clean-Up
rm(chelsea0711,team_results)
```

```r
### Functions for computing derivatives to determine values of constants in linear
### gaussian approximation of g(y_t | z_t ; psi)
# S(m,lambda)
S <- function(m,lam,goals){
  # lam: three-dim vector containing current attack/defense strengths, dep. parameter
  # goals: two-dim vector of goals in current week, used to calculate minimum
  Xt <- goals[1]
  Yt <- goals[2]
  k <- seq(0,min(goals))
  sum(choose(Xt,k) * choose(Yt,k)* factorial(k) * k^m * (lam[3]/lam[1]*lam[2])^k)
}

# U(m,lambda)
U <- function(m,lam_t,goals){
  S(m,lam_t,goals) / S(0,lam_t,goals)
}

# Udot(lambda)
Udot <- function(lam_t,goals){
  U(m=2,lam_t,goals)  - U(m=1,lam_t,goals)^2
}

# pdot(lam_t)
pdot_lam <- function(lam_t,goals){
  Xt <- goals[1]
  Yt <- goals[2]
  correction <- U(1,lam_t,goals)
  c(1/lam_t[1]*(Xt-lam_t[1]-correction),
    1/lam_t[2]*(Yt-lam_t[2]-correction))
}

# pddot(lam_t)
pddot_lam <- function(lam_t,goals){
  matrix(data=c(-1/lam_t[1]*
                  (1+ pdot_lam(lam_t,goals)[1] - 1/lam_t[1] * Udot(lam_t,goals)),
                1/lam_t[1] * 1/lam_t[2] * Udot(lam_t,goals),
                1/lam_t[1] * 1/lam_t[2] * Udot(lam_t,goals),
                -1/lam_t[2]*
                  (1+pdot_lam(lam_t,goals)[2] - 1/lam_t[2] * Udot(lam_t,goals))),
```

```r
         2,2)
}

# sdot(z_t)
sdot <- function(z_t,delta){
  # z_t contains current alpha and beta
  matrix(c(exp(delta+z_t[1]-z_t[2]),
           -exp(delta+z_t[1]-z_t[2]),
           -exp(delta+z_t[1]-z_t[2]),
           exp(delta+z_t[1]-z_t[2])),
         2,2)
}

# pdot(z_t) (to be summed over blocks of 10 matches)
pdot_z <- function(z_t,lam_t,delta,goals){sum(sdot(z_t,delta) %*% pdot_lam(lam_t,goals))}

# pddot(z_t) (to be summed over blocks of 10 matches)
pddot_z <- function(z_t,lam_t,delta,goals){
  sum(t(sdot(z_t,delta)) %*% pddot_lam(lam_t,goals) %*% sdot(z_t,delta))}

# g(y_t | z_t ; psi)
linear_gaussian <- function(lam_t,goals,psi,z_t,n){
  # psi: c(phi_alpha, phi_beta, sig2_alpha, sig2_beta, delta, gamma)
  # z_t: c(alpha_t, beta_t)
  # n: samples to generate
  phi_alpha <- psi[1]
  phi_beta <- psi[2]
  sig2_alpha <- psi[3]
  sig2_beta <- psi[4]
  delta <- psi[5]
  gamma <- psi[6]
  W <- c(1,-1)
  c_t <- goals - c(delta, 0) -
    t(W) %*% (
      z_t+ solve(pddot_z(z_t,lam_t,delta,goals)) %*% pdot_z(z_t,delta,goals))
  V_t <- t(W) %*% solve(pddot_z(z_t,delta,goals)) %*% W
  home <- rnorm(n,delta + z_t[1]-z_t[2] + c_t[1],sd=sqrt(V_t))
  away <- rnorm(n,z_t[1]-z_t[2] + c_t[2],sd=sqrt(V_t))
  # product of this needs to be taken...
  c(home,away)
}

### Candidate (taking form of likelihood)
approx_model <- function(z_t,lam_t,psi,goals,n){
  # g(z;psi)*g(y_t|z_t; psi)
  # given the previous time, g(z;psi) ~ N(mu+Phi*z_{t-1},H); normal density
  H <- matrix(c(psi[3],0,
                0,psi[4]),
              2,2,byrow=F)
  mu <- 0 # mu is another unknown constant? they don't really talk much about it
  dnorm(z_t,mean=mu+psi[1]*z_t[1]+psi[2]*z_t[2],H) %*%
    prod(linear_gaussian(lam_t,goals,psi,z_t))
}
```

```r
y <- matrix(c(epl_train$FTHG,epl_train$FTAG),ncol=2) # data matrix
start_ind <- seq(1,nrow(y),by=10) # starting index for each week, subset goals

### Initial values:
### -assume "average attack/ defense strength for z_0
### -assume home and away teams score at same rate for lam_0
### -assume no change over time, small disturbances, slight home-field advantage,
### slight goal-scoring covariance for
### psi = (phi_alpha, phi_beta, sig2_alpha, sig2_beta, delta, gamma)
z_0 <- c(0,0)
lam_0 <- c(1,1)
psi_0 <- c(1,1,0.01,0.01,0.1,0.1)

###  Save values
zs <- matrix(0,nrow=length(start_ind),ncol=2)
lams <- matrix(0,nrow=length(start_ind),ncol=2)
psis <- matrix(0,nrow=length(start_ind),ncol=6) # first two components main interest

approx_model(z_0,lam_0,psi_0,y[start_ind[1],])

### Process:
# approximate integral (likelihood) with monte carlo (importance sampling),
# take functional average maximize with importance sampling

### Importance sampling
IS_func <- function(f, g, n=10000, M=50){
  # f: target, g: candidate, n: desired number of samples, M: iterations
  # test with 50, 200, 1000 iterations
  for (i in 1:iter){
    samples <- approx_model(z_0,lam_0,psi_0,y[start_ind[1],],n)
    # sample psis from approximate model
    weights <- sapply(samples, function(psi) l(psi) / approx_model(psi))
    weights_normalized <- weights / sum(weights)
    psi_IS <- sum(samples^2 * weights_normalized)
    psis[i,] <- psi_IS
  }
}
colMeans(psis) # parameter estimates, should be close to Table 1
```

```r
### Maximization of Estimated likelihood
# From lecture notes
Gradmat<-function(x, f, eps = 1e-06){
  # Calculate the difference-quotient approx gradient (matrix) of an arbitrary
  # input (vector) function "f" at "x"
  vars = length(x) # number of variables to differentiate wrt
  aa = length(f(x)) # length of output
  eps_matrix = (diag(vars) * eps)/2
  gradient_matrix = array(0, dim = c(aa, vars))
  for(i in 1:var){
    # compute gradient for each variable
    gradiant_matrix[, i] <- (f(x + epsmat[, i]) -
    f(x - eps_matrix[, i]))/eps}
  if(aa > 1) gradient_matrix else c(gradient_matrix)
```

```r
}

MV_Newt_Raph <- function(guess,f, max.iter=25, tol = 1e-05, eps=1e-06){
  # guess: vector of initial values, changes from week to week
  # f: likelihood
  gradient <- function(x) Gradmat(x, f, eps)
  hessian <- function(x) Gradmat(x, gradient, eps)
  old <- guess
  new <- old - solve(hessian(old), gradient(old))
  it <- 1
  while(it < max.iter & sqrt(sum((new - old)^2)) >tol){
    old <- new
    new <- old - solve(hessian(old), gradient(old))
    it <- it + 1
  }
  return(new)
}

MV_Newt_Raph(psi_0,approx_model)
```

```r
### Individual teams
arsenal <- team_results$Data[[1]][,c(1:3,16,17)] # select dates and goals scored
# this is different from the author's implementation - they looked at all home
# goals and away goals for each week, rather than a specific team. this would
# just look at how a team's scoring/defense evolves over time REGARDLESS of
# whether they were home or away
arsenal$Team_Goals %>% arima(order=c(1,0,0))


### Grouping by week (extract attack/defense strength over time)
for (i in 2:length(start_ind)){
  matches <- y[start_ind[i]:(start_ind[i]+9),]
}
```

```r
# use rbp() with fitted values to generate, compare to actual distribution of results
# --> add third column that determines result, (apply ifelse vectorize)
# then tabulate each column, compare with frequencies
result_determiner <- function(match){
  if (match[1] > match[2]) match$Result <- 3
  else if (match[1] == match[2]) match$Result <- 1
  else if (match[1] < match[2]) match$Result <- 0
}

# using fitted values from their paper
pred1 <- rbp(38,lambda=c(1.5,0.9,0.0966)) %>% data.frame()
colnames(pred1) <- c("Scored", "Conceded")
pred1 <- mutate(pred1,result=apply(pred1,1,result_determiner))
# ^ half of this would be the average team's home results, the other half their away
pred_frequencies <- apply(pred1,2,table)
pred_frequencies$result <- pred_frequencies$result/nrow(pred1)
pred_frequencies

# rbp() can also be used to generate synthetic data, after which model fitting
```

```
# can be repeated
```

Plots

The first two plots below are an example of the output from the match_results function. The function (or the plots) could be easily altered to "zoom in" on specific results (time frames) if desired. The plots that follow are recreations of the goal plots in the original paper, but colorized to show the points gained by the team from the match. Note that red lines indicate losses (i.e. 0 points gained in the competition), green lines a draw (1 point gained), and blue lines a win (3 points gained). These plots can be used to visualize potential trends, along with sequences where the team won/drew/lost several matches consecutively. Lines appearing very thick suggests the team equaled their goal-scoring or conceding output in consecutive games.

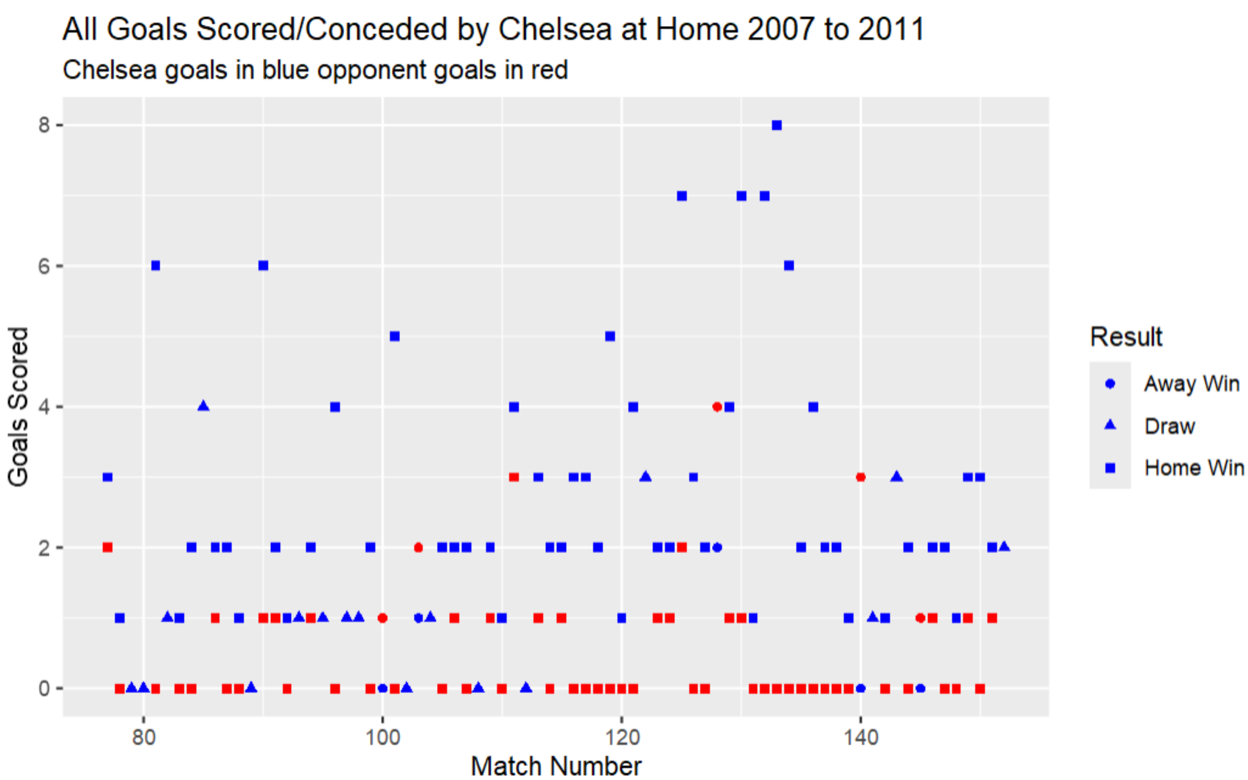

Figure 4: Chelsea Home Results 2007-2011

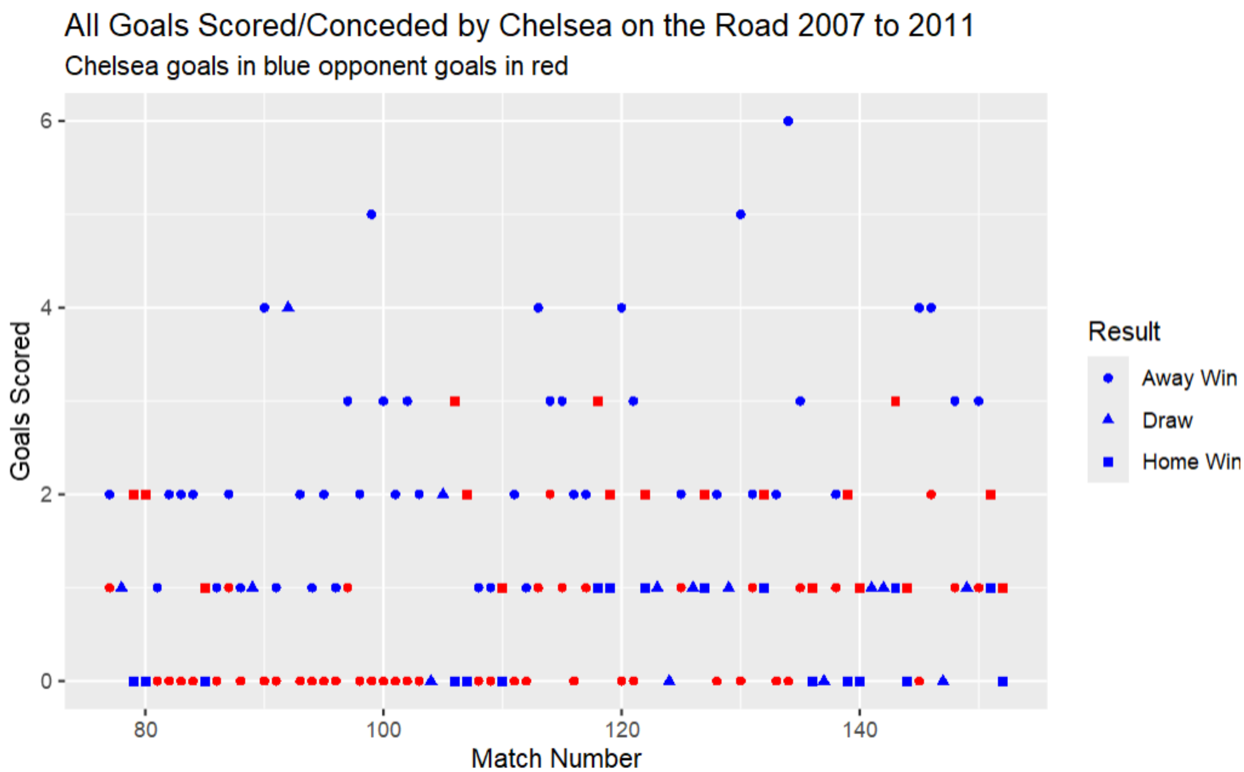

Figure 5: Chelsea Home Results 2007-2011

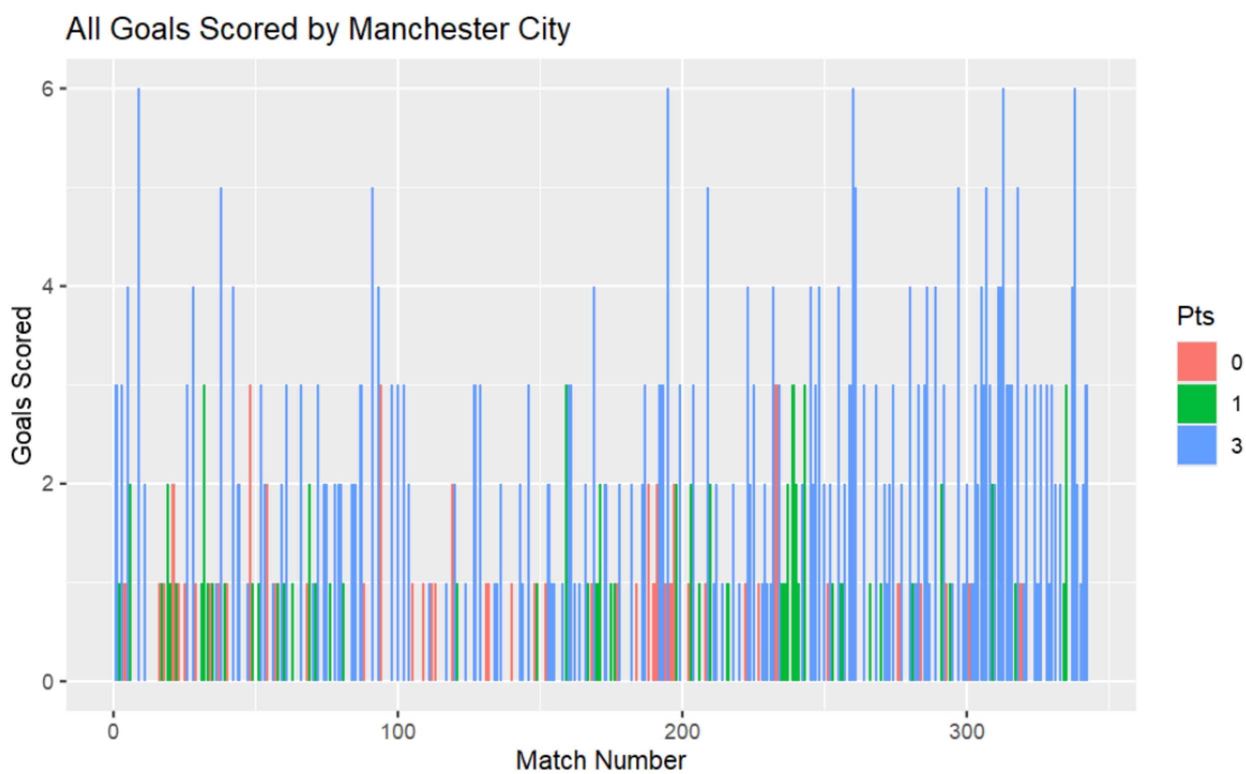

Figure 6: Manchester City Goals Scored

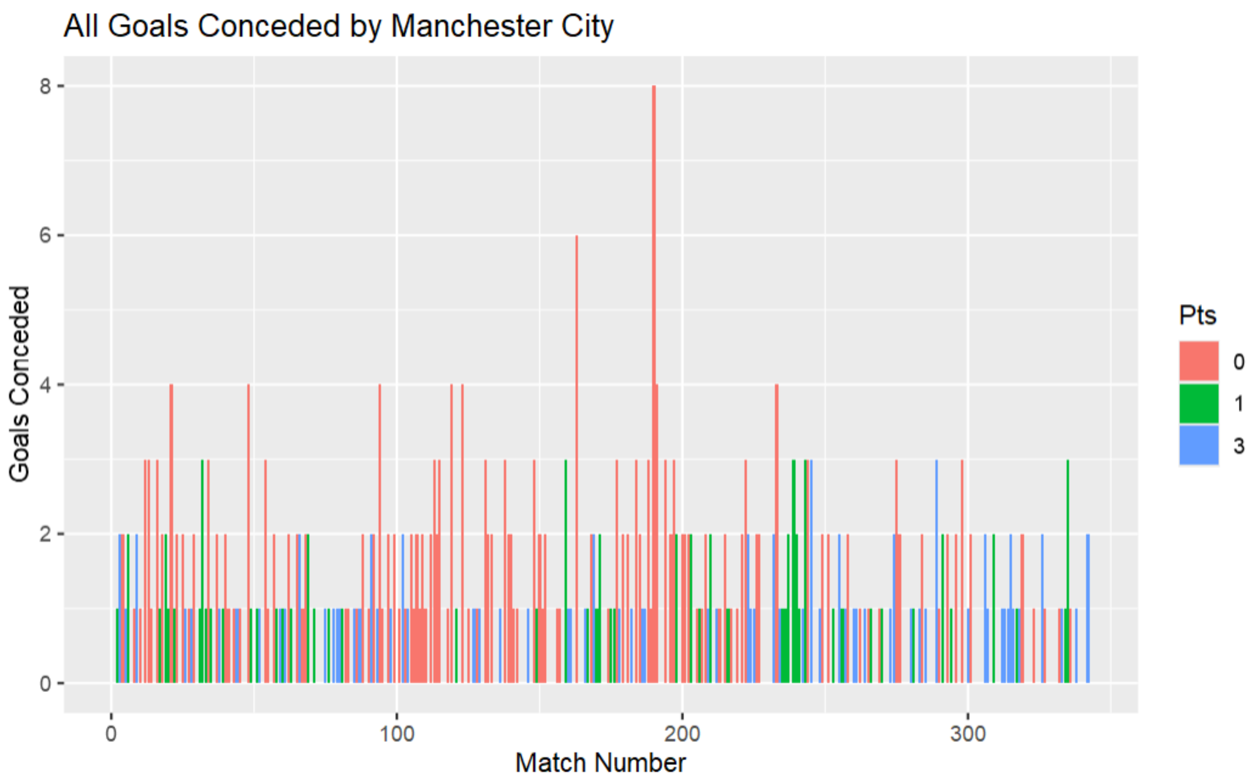

Figure 7: Manchester City Goals Conceded

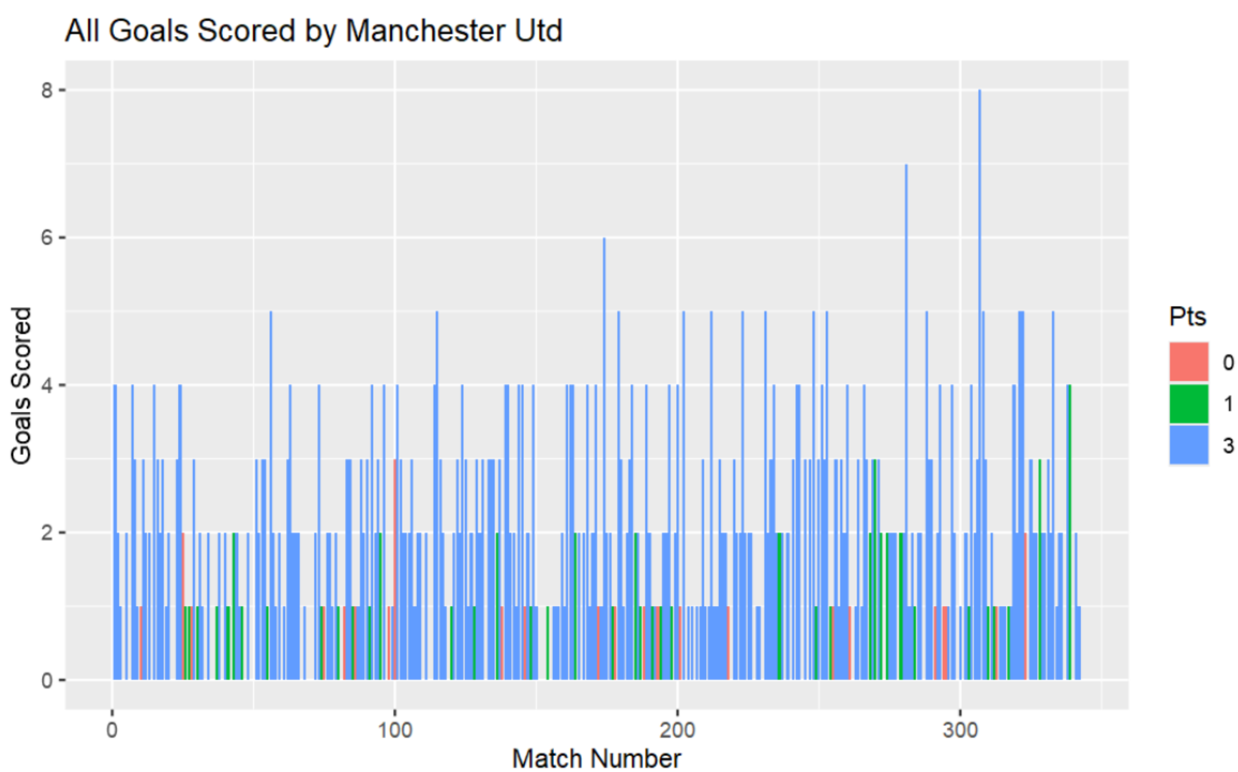

Figure 8: Manchester United Goals Scored

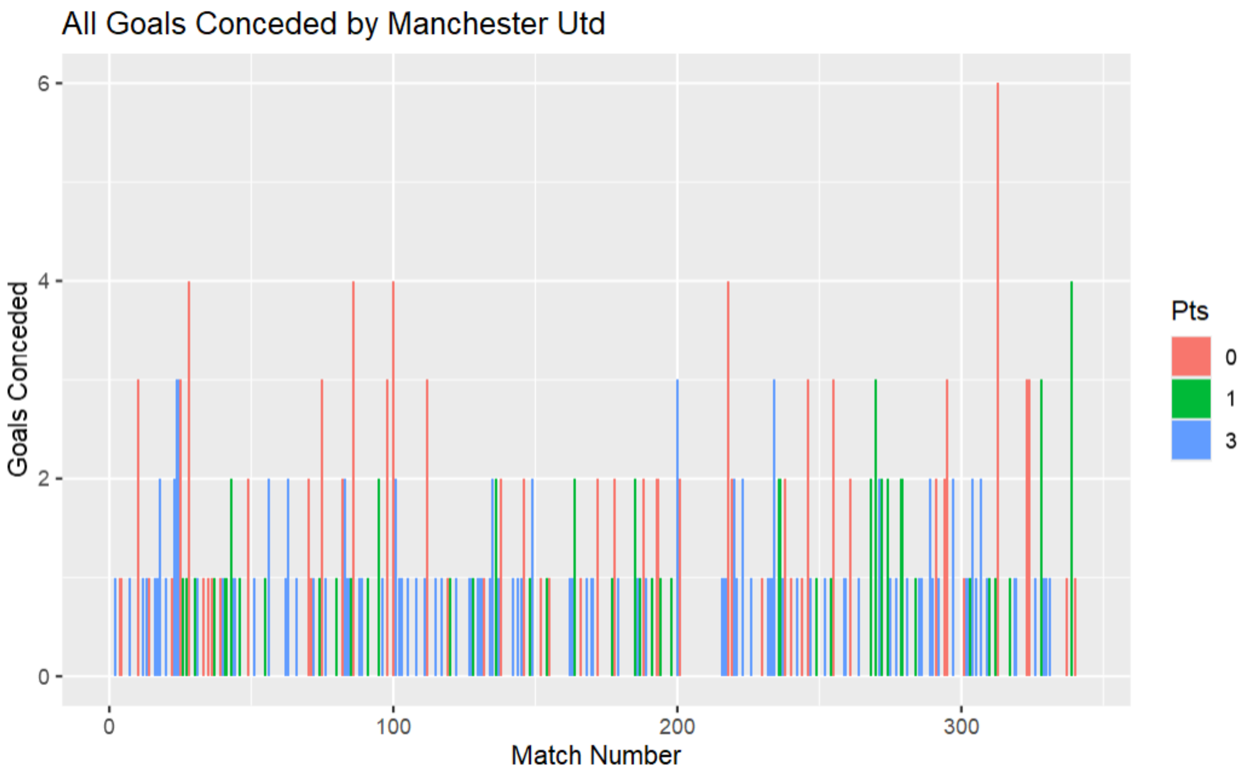

Figure 9: Manchester United Goals Conceded